EVAN YAP ZHI KAI

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EDUCATION

SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN **Singapore**

Bachelor of Engineering (Engineering Systems and Design), Masters of Science in Technology Entrepreneurship *Expected May 2030*

Relevant Coursework: Algorithms & Object-Oriented Programming, Machine Learning, Data Modelling & Visualisation

- **1 of the 6 recipients** of the SUTD Technology Entrepreneurship Programme Scholarship

PROFESSIONAL EXPERIENCE

Univers Pte Ltd | ***Intern - AI Business Consultant (Aviation, Transportation, Logistics - GTM)*** *Aug 2026 - Present*

- Built and deployed an AI Decarbonization & Audit Agent (Next.js, Vercel AI SDK, Groq LPU inference) to automate ESG disclosure ingestion and client claim verification, cutting manual research time by ~3 hours per facility audit.
- Developed and calibrated an HVAC energy-consumption regression model to validate client energy assumptions, achieving estimates within 2-3 percentage points of observed consumption ranges and identifying key efficiency opportunities.
- Analysed operational workflows, industry benchmarks, and stakeholder requirements to define and prioritise AI and energy-management use cases by expected impact, telemetry availability, technical feasibility, and implementation effort.
- Advised the Senior Director, Global Lead, and senior client stakeholders by translating technical analysis into decision-ready business cases, prioritised solution requirements, decarbonisation recommendations, and implementation roadmaps.

EvanYapFitness | ***Founder*** *Sept 2024 - Jan 2026*

- Founded and scaled an online coaching brand, reaching an audience of 2.2k followers and 10k average views per post by iterating on content performance analytics and A/B-tested messaging.
- Designed and implemented a structured intake pipeline using Google Form and Excel to capture training history, lifestyle constraints, and nutrition habits for 20+ clients, enabling reproducible and scalable programming.
- Deployed an AI-assisted recommendation workflow using Perplexity to generate training and meal templates from client profiles, which were then refined manually, cutting average programming time per client by 50%.

RELEVANT PROJECTS

Univers Website Revamp Challenge (Top Winner) | Vanilla JavaScript (ES6+), Three.js (WebGL), CSS3, Web Audio API, Vercel

Frontend Developer *Sept 2026*

Beat global senior solution architects to reach the **Top 5**, with a 9-page interactive redesign of Univers' EnOS™ Physical AI platform website.

- Pitched the redesign live to Univers' CEO and senior executives, explaining design decisions in business terms (brand alignment, enterprise lead generation, platform storytelling) and answering their technical and strategic questions.
- Built the site with no framework and no build step, in modular vanilla JavaScript and CSS, including a custom i18n engine serving 7 languages with each visitor's language choice saved between visits.
- Engineered a real-time WebGL particle mesh in Three.js that reacts to the cursor with spring physics and to scrolling with momentum, lazy-loaded via dynamic import() so the site still works without it.
- Shipped an executive ROI modeler for uptime, energy-cost and carbon savings, a 3-stage edge-to-cloud operations simulator, and odometer-style telemetry counters triggered by IntersectionObserver.
- Made it production-ready with preferred-reduced-motion fallbacks, semantic accessible markup, immutable asset caching, and a Vercel serverless demo-booking API deployed automatically on every push.

Personal Assistant Bot - Independent Project | Next.js, TypeScript, Vercel AI SDK, Gemini, Perplexity Sonar, Google Cloud Pub/Sub, Gmail/Sheets/Calendar APIs, Zod, Vercel

Full-Stack Developer *Aug 2026 - Sept 2026*

Serverless Telegram assistant handling 24 natural-language commands via text, voice or photo to automate calendars, expense tracking, bill splitting and email.

- Built payment ingestion for Apple Pay (iOS Shortcuts), DBS/PayLah!, Grab and e-commerce receipts via Gmail Pub/Sub push; parallel parsing with Promise.allSettled and cut sync time by ~77%.
- Designed a 3-tier AI intent engine (Gemini Flash-Lite → 3.6 Flash → Perplexity Sonar failover) routing 24 Zod-validated intents at 100% eval accuracy and ~65% lower cost per call.
- Built a multimodal pipeline that transcribes voice notes and extracts up to 30 events or 25 receipt items per image, including bill splitting, live FX conversion and free-slot finding.
- Hardened security with constant-time auth, prompt-injection guardrails, PII redaction and AES-256-GCM encryption; persisted undo tokens and idempotency logs across serverless cold starts.

KEY SKILLS

Languages: Python, TypeScript, JavaScript, C++, C, SQL, HTML/CSS

Frameworks & Technologies: Next.js, React, Node.js, Vercel AI SDK, REST APIs, Webhooks, Zod, OAuth 2.0, Telegram Bot API

Cloud & Developer Tools: Vercel, GitHub, Google Cloud APIs, API integration, Environment variables

Leadership: SUTD Men's Floorball Captain, SUTD House Executive Committee (Floor Events & Hostel Works Rep)

ADDITIONAL INFORMATION

Interests: Competitive Fitness & Powerlifting (HYROX Athlete), Playing the guitar, Coffee, Volunteer work (Church Worship Ministry)